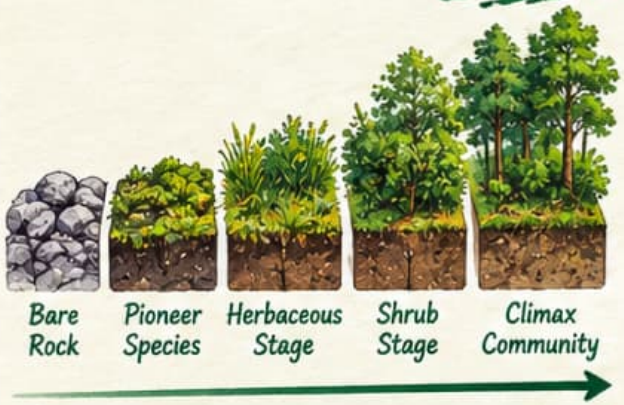
# ABN SEAL COLLEGE
## COOCHBEHAR
### PANCHANAN BARMA UNIVERSITY
### COOCHBEHAR

# Dynamics and Succession




## B.Sc. 2nd Semester

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ø What is succession? Describe different steps involved in the process of succession. 2+8

> Succession- It is the act or process of following in order.

Ecological succession-

> Ecological succession is defined as the non-seasonal, directional and continuous pattern of colonization and coextinction ona site by species population.

> It involves changes in energy partitioning, species structure and community ~~pos~~ processes over time.

> The whole sequence of communities that replace one another in a given area is termed as sere, and the relatively transitory communities during succession are termed as seral stages or developmental stages.

> The initial sere is termed pioneer ~~spec~~ ~~stat~~ stage while the terminal stage or mature stabilized system is the climax stage.

> pioneer community ⟶ intermediate community ⟶ climax community.

General process of succession-

ø ~~Nudation~~ In 1916, Frederic clements proposed the theory of succession which is usually termed as 'classical ecological theory'. According to him succession is a process ~~of~~ involving several sequential phases that follow one another. The phases are-

① Nudation-

Nudation is the formation or development of a bare area without any form of life where succession begins. The area may develop due to several causes-

ⓐ Topographic : soil erosion, landslide, volcanic activity earthquake etc.

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- Climatic : Glaciers, frost, fire, strom.
- Biotic : Deforestation, urbanization, industrialization by distruction of forest, disease due to ~~fi~~ fungi, bacteria etc.

② Invasion:—

Successful establishment of species in a bare area. The species actually reaches the bare area from any other site. It involve following successive stages-

a) Migration (dispersal): The seeds, spores of the new species reach the bare area by the help of wind, water animal including humans.

b) ~~Ecesis~~ Ecesis (establishment): After reaching the area the process of ~~succe~~ successful establishment of the species ~~is~~ is known as ecesis.

→ The process depends on climatic, edaphic & biotic factors.

→ After migration seeds or propagulates germinate, seeding grow & adult start to reproduce.

→ Only few of them can survive & grow under such primitive & harsh environment, most of them disappear.

c) Aggregation: As a result of reproduction, the individuals of the species increase in number and they come closer to each other.

③ Competition & Coaction-

It is phenomenon involve struggle for existance between two or more individuals growing in an area, that makes successive demands, that are ~~simila~~ similar in nature, on the soil. The struggle is usually between two individuals of same kind that have similar demands such as space, nutrients, sunlight, water etc. Competition can be either interspecific (b/w two different species) or intraspecific (within the same species). As a result superior species eliminates inferior one. The dead body of plants &

animals of eliminated species decay & form humus which enrich the soil.

④ Reaction - This is the most important stage of succession. The mechanism of the modification of the environment through the influence of living organisms on it, is known as reaction. As a result of reaction, organism changes the soil, water, temperature etc of the environment. For example plant changes the soil structure & texture by adding humus into the soil. Due to all these ~~soil~~, environment is modified, becoming unsustainable for the existing community & sooner or later is replaced by another community.

⑤ Climax community - (Stabilization)

The reaction of individuals collectively have a wider effect on the environment. The local ~~community~~ climate is actually changed if the climax vegetation is forest. This community is relatively more stable. Thus, we found the climax community which is the ~~rest~~ result of its reactions.

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Ø Discuss briefly about hydrosere / Hydrarch succession.

> Plant succession which begins in ponds, lakes, marshes or elsewhere in water are termed as hydrarch and different stages are called hydrosere. (Remaining xerox)

Ø Discuss briefly about xerarch succession

>Xerosere is the sequence of successional state which occur on bare rock/land deficient in water. (Remaining xerox)

Ø Climax community.

A climax community is the final, stable stage of succession where the ecosystem is balanced & self sustaining. It does not change much over time unless disturbed.

example-

Tropical rainforest in a humid region or Grassland in dry areas.

Ø Climax community-

A climax community is the final, stable stage of ecological succession in an ecosystem. It is a mature community that remains in equilibrium with the environment for a long period unless disturbed by natural desaster or ~~by~~ human activities.

Characteristics of Climax community-

① Stable & self sustaining- The community maintanes balance with the climate & soil conditions.

② High Biodiversity- It contains a large variety of plants and animals.

③ Complex food web- Organisms are interconnected through complex feeding relationships.

④ Efficient nutrient cycling- Nutrients are recycled effectively within the ecosystem.

⑤ Dominance & Long-lived species- Large trees & perennial plants are common in terrestial climax communities.

⑥ Resistance to Environmental changes- It can tolerate minor environmental fluctuations.

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examples –

> A dense tropical rainforest in a humid regions.
> A mature deciduous forest in temperate areas.
> Stable grassland ecosystems in dry regions.

Importance –

> Maintain ecological balance.
> provides habitat for numerous organisms.
> prevents soil erosion & conserves nutrients.
> Supports long term ecosystem stability.

Thus, a climax community represents the most developed & balanced stage of ~~ecos~~ ecological succession.

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| Basis | Primary succession | Secondary succession |
|---|---|---|
| 1) Defn | Succession starts from a bare area where no life exists. | Succession starts from an area where a community previously existed but was disturbed. |
| 2) presence of soil | soil is absent at the beginning | soil is already present. |
| 3) Starting area. | Begins on bare rocks, lava, sand dunes or newly formed land. | Begins on burned forest abondoned field, flood-affected areas etc. |
| 4) pioneer species | Lichens, mosses are usually pioneer species. | Grasses, humb, shrubs are usually pioneer species |
| 5) Rate of succession | very slow process because soil formation takes time. | Comparatively faster because soil & organic matter already exist. |
| 6) previous life. | No previous biological community existed. | previous community existed before distruction or distrurbance |
| 7) Time reqd to reach Climax community | Takes houndreds or thousands of years to reach climax community. | Takes comparatively less time often decades or centuries. |
| 8) ex. | Colonization of bare rock | Regrowth of vegetation after a forest fire / flood. |

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Ø Types of Ecological succession.

Plant succession can be defined as the process of gradual replacement of one plant community by another plant community which is stable type. It occurs over a period of time

Here are the some types of succession -

1) primary Succession -

This type of succession occurs in areas where there was no ~~life~~ previous life, such as newly formed rocks after volcanic erruptions etc. The process starts from completely barren and takes a long time as soil has to form first. pioneer species like lichens & mosses are the first to colonize. ex- lava flows, cools and solidify creating a new area where lichens & mosses first colonize.

2) Secondary succession -

Secondary succession occurs in areas where a previou community existed but was destroyed by events like fire, flood or human activities. Since the soil is already present the area recover faster with plants & Animals gradually recolonizing & restoring the ecosystem.

ex- ~~After a~~ fire grasses & shrubs may quickly regrow, followed by trees leading to forest.

3) Autogenic succession -

Autogenic succession is when existing community modifies its own environment through biological activities such as changing soil quality or light availability. These changes makes the habitat suitable for new ~~the~~ species leading to gradual replacement of original community by another.

ex- Formation of mature forest from abondoned agri-field.

4) Allogenic succession -

Allogenic succession occurs when changes in a community happen due to external factors like erosion, pollution or climate changes, rather than the organisms theirself modifying their environment / habitat. Those outside forces cause the replacement ~~one~~ of one community by the other.

ex- Forest fire clear the existing vegetation, allowing for the establishment of new species that may be better adopted to the post environment.

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5) Autotrophic succession-

Autotrophic succession begins ~~with~~ in areas with low organic matter and is dominated by ~~plant~~ green plants. It leads to increase biomass through continuous energy flow from photosynthesis.

Here green plants and other ~~predominantly~~ photosynthetic organisms dominate the early stages. It begins predominantly inorganic environment.

6) Heterotrophic succession-

Heterotrophic succession starts in areas rich in organic matter and is initially dominated by organisms like bacteria & fungi. It usually leads to gradual decline in energy content over time.

It begins in a predominantly organic environment.

7) Induced succession-

It occurs due to human activities like deforestation, pollution etc. These action disturb stable ecosystem.

8) Cyclic succession-

It occurs when small disturbance create opening within a stable ~~an~~ ecosystem, leading to repeated reappearance of earlier successional stages.

9) progressive succession-

progressive succession drives an ecosystem towards more complexity, higher biomass & more mesic conditions

ex → Bare rock → lichens → mosses → Grasses → shrubs → A mature forest

10) Retrogressive succession-

It moves an ecosystem backword, causing biodiversity loss, decreased complexity & degradation.

ex → A mature forest ——turns——→ Grassland.

Ⓐ Add hydrarch & Xerarch succession in brief.

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## ø Climax theory –

### Concept of climax

> Climax is the final, stable stage of ecological succession where the ecosystem reaches balance with its environment.
eg → Forest, desert, grassland, coral reef.

> It is the final stage of succession, stable & balanced ecosystem, which has no major changes until or unless any disturbance.

> Succession leads to climax stage, ecosystem reaches equilibrium, controlled mainly by environment.

> Climax is not random, it follows a pattern.

### Climax Theory

> Climax theory states that ecological succession leads to a final stable stage called climax community which is in equilibrium with the environment and remains relatively unchanged unless disturbed.

> Ecosystem develops step by step through succession.

> From pioneer stage → intermediate ~~state~~ stage → climax stage

> The climax stage is self sustaining and stable.

### Types of Climax theory

Three major theories ~~are~~ have been proposed to define, identify & interpreter the climax communities. These theories are –

i) Monoclimax theory.
ii) Polyclimax theory.
iii) Climax pattern theory.

### i) Monoclimax theory –

> proposed by Frederic Clements

> Monoclimax theory states that – in a given region ecological succession always lead to one single, final climax community, mainly determined by climate.

> Says that only one climax community exists in a region.

> Controlled mainly by environment.

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Basic concept of ~~eli~~ monoclimax theory-

> Succession is orderly and predictable.
> All communities pass through stages
> ~~poi~~ pioneer → intermediate → climax.
> Regardless the starting condition they ~~rich~~ reach the same climax in that region.

> example →
in tropic region → climax → dense forest
in desert region → climax → desert ~~forest~~ vegetation.

Types-

1) Climatic climax - determined by climate
2) Sub climax - Stage just before climax
3) Disclimax - caused by disturbance
4) pre climax - occurs in slightly drier condition than climax
5) post climax - occurs in slightly wetter condition than climax

ii) Polyclimax theory-

> proposed by ~~Arthur~~ Arthur Tansley.
> polyclimax theory states that in a given region, there can be many climax communities, each controlled by biotic & different environmental factors, not just by climate.
> Says that many climax community exists in a region.
> Controlled mainly by biotic & environmental factors.

Basic concept of polyclimax theory-

> Succession does not always end in one single climax
> Different areas of same region may ~~have~~ develop different climax communities.
> Climax depend on multiple interacting factors, for example soil, climate, topography, fire, ~~etc~~ animal and human activities.

ex- forest in moist areas
~~des~~ Grassland in dry areas.

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Types-

① Climatic climax- controlled by climate
② Edaphic climax- Controlled by soil condition
③ Biotic climax- Controlled by organisms (humans, animals)
④ Fire climax- maintained due to frequent fires.
⑤ Topographic climax- Controlled by landforms; (slope, Altitude).

iii) Climax pattern theory-

> It is a variation of polyclimax theory proposed by R.H. Whittaker.

> The natural community is adaped ~~the~~ to the whole pattern of environmental factors in which it exists.

> Where monoclimax theory allows only one climatic ~~regio~~ climax ~~&~~ in a region & polyclimax theory allows several climaxes in a same area, climax pattern ~~is~~ pattern hypothesis allows a whole range of edaphic factors, fire & biotic factor, including grazing, diseases & ~~mutula~~ mutualistic & competitive interactions.

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ø Significance of ecological succession-

The significance of ecological succession lies in how it helps ecosystems develop, recover & maintain balance over time. The significance of ecological succession explained below-

① Development of ecosystem-

> Succession shows how a bare or disturbed area gradually ~~is~~ turns into a ~~et~~ complex ecosystem with plants, animals & microorganisms.

② Soil formation & improvement -

> pioneer species help in soil formation, later stages ~~impt~~ improve soil fertility.

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3) Increase in Biodiversity-

> Number & variety of species increase over time leads to a complex food ~~chain~~ chain & food web that increase the biodiversity.

4) Nutrient cycling-

> Succession establishes proper nutrient cycling like carbon cycle, nitrogen cycle which makes the ecosystem self sustaining.

5) Ecosystem recovery-

helps in natural restoration after disturbance such as-

- forest fires
- floods
- human activities

6) Stability & balance-

> leads to formation of a climax community.

> ecosystem became stable & balanced.

7) Climate regulation-

> Succession helps in developing vegetation that can regulate local climate by controlling temperature, humidity, and carbon dioxide levels.

8) Habitat formation-

> It creates suitable habitats for different organism over time, supporting wildlife & increase ecological diversity.

9) prevention of soil erosion-

> As vegetation increase it helps in binding soil, thereby reducing soil erosion caused by wind & air.

10) Climax community formation-

> By the process of succession a stable & mature ecosystem forms. The ecosystem remains balanced over time.

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